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Effect of unequal injection rates and different hopping rates on asymmetric exclusion processes with junction*

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In this paper, the effects of unequal injection rates and different hopping rates on the asymmetric simple exclusion process (ASEP) with a 2-input 1-output junction are studied by using a simple mean-field approach and extensive computer simulations. The steady-state particle currents, the density profiles, and the phase diagrams are obtained. It is shown that with unequal injection rates and different hopping rates, the phase diagram structure is qualitatively changed. The theoretical calculations are in good agreement with Monte Carlo simulations.

Keywords: asymmetric simple exclusion processes, mean-field approximation, junction, different hopping rates

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1. Introduction

Asymmetric simple exclusion process (ASEP) is a very prominent tool which has been widely used to study low-dimensional non-equilibrium systems in chemistry, physics and biology,[1−8] such as diffusion through membrane chains and traffic flow analysis.[9−11] The ASEP is a one-dimensional lattice model where particles hop along discrete lattices. When particles hop only along one direction, the ASEP is called the total ASEP (TASEP). In these models, each of the sites has only two states, i.e., empty and occupied by a single particle. Several exact solutions for the stationary properties of ASEP have been obtained such as those for periodic[12] and open boundary conditions[13] and different update rules.[14−16]

Recently, several more realistic phenomena have been analysed via the TASEP models, e.g. multi-channel lattices,[17] two-channel lattices,[18−24] lattices with a junction.[25−28] In Ref. [25], Pronina and Kolomeisky investigated the TASEP on lattice with junctions; and developed a simple approximate theory. More recently, the TASEP on lattice with a junction has also been studied by Wang *et al.*[28] In Ref. [28], different injection rates were considered. Several new stationary phases were found in the system. We note that the junction points can be viewed as bottlenecks which may lead to traffic jams. Therefore, the study of TASEP with a junction may have strong implications on complex biological transport, vehicular traffic even information packet transport.

In this paper, we study the effects of unequal injection rates and different hopping rates on TASEP with a 2-input 1-output junction. It is shown that with the unequal injection rates and different hopping rates, the phase diagram structure is qualitatively changed. The density profiles of the stationary phases and the phase boundaries are investigated in detail by using Monte Carlo simulations and mean-field approximation. The remainder of this paper is organized as follows. In Section 2, a detailed description of the model is given, and exact theoretical solutions are presented. In Section 3, we discuss Monte Carlo computer simulations and compare them with theoretical predictions. We give our summary and conclusions in Section 4.

2. Model and theoretical analysis

The schematic picture of the model is shown in Fig. 1(a). The system includes three chains, each of them includes L sites. Particles can inject sub-chain 1 and sub-chain 2 with rates α_1 and α_2, respectively, if there are no particles on the first site of two parallel sub-chains (i.e., sub-chain 1 and sub-chain 2). At the

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junction, two parallel sub-chains merge into a single chain (i.e., main-chain). Particles can eject the system with rate β if there is a particle on the last site of the main-chain. In different bulks, the hopping rates are different. The parameters p_1, p_2 and p_3 are used to express the hopping rates of sub-chain 1, sub-chain 2 and main-chain, respectively (see Fig. 1(b)). Firstly, we briefly review some results for the TASEP with reduced hopping rate and random update.[16]

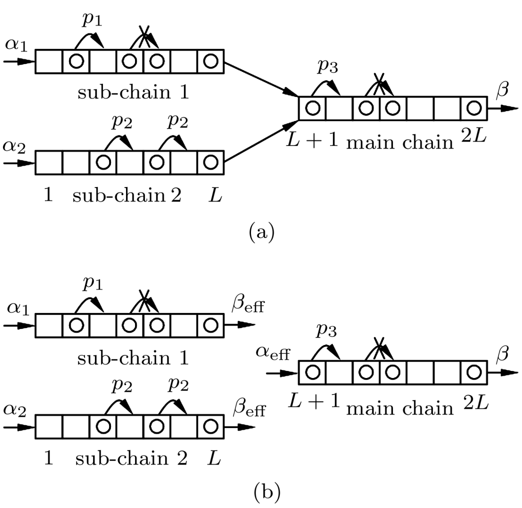


Fig. 1. Schematic diagrams of unequal injection rates and different hopping rates in TASEP with a junction (a) and effective injection and removal rates at junction points (b).

For $\alpha < \beta$ and $\alpha < p/2$, the system is in the low-density (LD) phase, for which

$$J = \alpha\left(1-\frac{\alpha}{p}\right), \quad \rho = \frac{\alpha}{p}, \tag{1}$$

where J is the system current and ρ is the bulk density. For $\beta < \alpha$ and $\beta < p/2$, the system is in the high-density (HD) phase, for which

$$J = \beta\left(1-\frac{\beta}{p}\right), \quad \rho = 1-\frac{\beta}{p}. \tag{2}$$

For $\alpha > p/2$ and $\beta > p/2$, the system is in the maximal current (MC) phase, for which

$$J = p/4, \quad \rho = 1/2. \tag{3}$$

Due to the symmetry of sub-chains 1 and 2, we study the case where $\alpha_1 > \alpha_2$ (i.e., $\alpha_2 = n\alpha_1$ and $n < 1$). Next we discuss the five possible phases in the system.

The HD/HD/MC phase is determined by the following general conditions:

$$\begin{aligned}&\alpha_1 > \beta_{\rm eff}, \quad \beta_{\rm eff} < \frac{p_1}{2}, \quad \alpha_2 > \beta_{\rm eff},\\ &\beta_{\rm eff} < \frac{p_2}{2}; \quad \alpha_{\rm eff} > \frac{p_3}{2}, \quad \beta > \frac{p_3}{2}.\end{aligned} \tag{4}$$

According to Eqs. (2) and (3) and current conservation, we obtain

$$\beta_{\rm eff} = \frac{2p_1p_2 - \sqrt{4p_1^2p_2^2 - p_1p_2p_3\,(p_1+p_2)}}{2\,(p_1+p_2)}. \tag{5}$$

Therefore, for the (HD/HD/MC) phase to exist, one has

$$\begin{aligned}&\alpha_2 > \frac{2p_1p_2 - \sqrt{4p_1^2p_2^2 - p_1p_2p_3\,(p_1+p_2)}}{2\,(p_1+p_2)},\\ &\beta > \frac{p_3}{2}.\end{aligned} \tag{6}$$

Similar calculations can be performed for the HD/HD/HD phase, which exists when

$$\begin{aligned}&\alpha_1 > \beta_{\rm eff}, \quad \beta_{\rm eff} < \frac{p_1}{2}, \quad \alpha_2 > \beta_{\rm eff},\\ &\beta_{\rm eff} < \frac{p_2}{2}; \quad \alpha_{\rm eff} > \beta, \quad \beta < \frac{p_3}{2}.\end{aligned} \tag{7}$$

According to Eq. (2) and current conservation, we have

$$\beta_{\rm eff} = \frac{p_1p_2p_3 - \sqrt{p_1^2p_2^2p_3^2 - p_1p_2p_3\,(p_1+p_2)\,\beta\,(p_3-\beta)}}{p_3\,(p_1+p_2)}. \tag{8}$$

Thus, the system is in (HD/HD/HD) phase when

$$\begin{aligned}&\frac{p_1p_2p_3 - \sqrt{p_1^2p_2^2p_3^2 - p_1p_2p_3\,(p_1+p_2)\,\beta\,(p_3-\beta)}}{p_3\,(p_1+p_2)} < \alpha_2,\\ &\beta < \frac{p_3\,(p_1+p_2) - \sqrt{p_3^2\,(p_1+p_2)^2 - 4p_1p_2p_3\,(p_1+p_2)}}{2\,(p_1+p_2)}.\end{aligned} \tag{9}$$

The case with the HD/LD/MC phase can be generally described by the following conditions:

$$\alpha_1 > \beta_{\rm eff}, \quad \beta_{\rm eff} < \frac{p_1}{2}, \quad \alpha_2 < \beta_{\rm eff}, \quad \alpha_2 < \frac{p_2}{2}; \quad \alpha_{\rm eff} > \frac{p_3}{2}, \quad \beta > \frac{p_3}{2}. \tag{10}$$

According to Eqs. (1), (2), (3) and current conservation, one has

$$\beta_{\rm eff}=\frac{p_1p_2-\sqrt{p_1^2p_2^2-p_1p_2^2p_3+4p_1p_2\alpha_2\left(p_2-\alpha_2\right)}}{2p_2}. \tag{11}$$

Thus, the conditions for the (HD/LD/MC) phase to exist are

$$\begin{aligned}&\alpha_1>\frac{p_1p_2-\sqrt{p_1^2p_2^2-p_1p_2^2p_3+4p_1p_2\alpha_2\left(p_2-\alpha_2\right)}}{2p_2},\\&\alpha_2<\frac{2p_1p_2-\sqrt{4p_1^2p_2^2-p_1p_2p_3\left(p_1+p_2\right)}}{2\left(p_1+p_2\right)},\quad \beta>\frac{p_3}{2}.\end{aligned} \tag{12}$$

The case with the HD/LD/HD can be generally described by the following conditions:

$$\alpha_1>\beta_{\rm eff},\quad \beta_{\rm eff}<\frac{p_1}{2},\quad \alpha_2<\beta_{\rm eff},\quad \alpha_2<\frac{p_2}{2};\quad \alpha_{\rm eff}>\beta,\quad \beta<\frac{p_3}{2}. \tag{13}$$

According to Eqs. (1), (2) and current conservation, one has

$$\beta_{\rm eff}=\frac{p_1p_2p_3-\sqrt{p_1^2p_2^2p_3^2-4p_1p_2p_3\left[p_2\beta\left(p_3-\beta\right)-p_3\alpha_2\left(p_2-\alpha_2\right)\right]}}{2p_2p_3}. \tag{14}$$

As a result, the (HD/LD/HD) phase can exist when

$$\begin{aligned}&\alpha_1>\frac{p_1p_2p_3-\sqrt{p_1^2p_2^2p_3^2-4p_1p_2p_3\left[p_2\beta\left(p_3-\beta\right)-p_3\alpha_2\left(p_2-\alpha_2\right)\right]}}{2p_2p_3},\\&\alpha_2<\frac{p_1p_2p_3-\sqrt{p_1^2p_2^2p_3^2-p_1p_2p_3\left(p_1+p_2\right)\beta\left(p_3-\beta\right)}}{p_3\left(p_1+p_2\right)},\quad \beta<\frac{p_3}{2}.\end{aligned} \tag{15}$$

For the LD/LD/LD phase, it exists when

$$\alpha_1<\beta_{\rm eff},\quad \alpha_1<\frac{p_1}{2},\quad \alpha_2<\beta_{\rm eff},\quad \alpha_2<\frac{p_2}{2};\quad \alpha_{\rm eff}<\beta,\quad \alpha_{\rm eff}<\frac{p_3}{2}. \tag{16}$$

According to Eq. (1) and current conservation, we can obtain

$$\alpha_{\rm eff}=\frac{p_1p_2p_3-\sqrt{p_1^2p_2^2p_3^2-4p_1p_2p_3\left[p_2\alpha_1\left(p_1-\alpha_1\right)+p_1\alpha_2\left(p_2-\alpha_2\right)\right]}}{2p_1p_2}. \tag{17}$$

Thus, the conditions for the (LD/LD/LD) phase to exist are

$$\frac{p_1p_2p_3-\sqrt{p_1^2p_2^2p_3^2-4p_1p_2p_3\left[p_2\alpha_1\left(p_1-\alpha_1\right)+p_1\alpha_2\left(p_2-\alpha_2\right)\right]}}{2p_1p_2}<\min\left(\beta,\frac{p_3}{2}\right). \tag{18}$$

By taking into account $\alpha_2=n\alpha_1$, we can obtain the following results:
for the (HD/HD/MC) phase,

$$\alpha_1>\frac{2p_1p_2-\sqrt{4p_1^2p_2^2-p_1p_2p_3\left(p_1+p_2\right)}}{2n\left(p_1+p_2\right)},\quad \beta>\frac{p_3}{2}; \tag{19}$$

for the (HD/HD/HD) phase,

$$\frac{p_1p_2p_3-\sqrt{p_1^2p_2^2p_3^2-p_1p_2p_3\left(p_1+p_2\right)\beta\left(p_3-\beta\right)}}{np_3\left(p_1+p_2\right)}<\alpha_1,\quad \beta<\frac{p_3}{2}; \tag{20}$$

for the (HD/LD/MC) phase,

$$\begin{aligned}&\frac{p_1p_2\left(1+n\right)-\sqrt{p_1^2p_2^2\left(1+n\right)^2-p_1p_2p_3\left(p_2+p_1n^2\right)}}{2\left(p_2+p_1n^2\right)}<\alpha_1,\\&\alpha_1<\frac{2p_1p_2-\sqrt{4p_1^2p_2^2-p_1p_2p_3\left(p_1+p_2\right)}}{2n\left(p_1+p_2\right)},\quad \beta>\frac{p_3}{2};\end{aligned} \tag{21}$$

for the (HD/LD/HD) phase,

$$\frac{p_1p_2p_3\left(1+n\right)-\sqrt{p_1^2p_2^2p_3^2\left(1+n\right)^2-4p_1p_2p_3\beta\left(p_3-\beta\right)\left(p_2+p_1n^2\right)}}{2p_3\left(p_2+p_1n^2\right)}<\alpha_1,$$
$$\alpha_1<\frac{p_1p_2p_3-\sqrt{p_1^2p_2^2p_3^2-p_1p_2p_3\left(p_1+p_2\right)\beta\left(p_3-\beta\right)}}{np_3\left(p_1+p_2\right)},\quad \beta<\frac{p_3}{2}; \tag{22}$$

for the (LD/LD/LD) phase,

$$\alpha_1<\frac{p_1p_2\left(1+n\right)-\sqrt{p_1^2p_2^2\left(1+n\right)^2-p_1p_2p_3\left(p_2+p_1n^2\right)}}{2\left(p_2+p_1n^2\right)},\quad \beta>\frac{p_3}{2};$$
$$\alpha_1<\frac{p_1p_2p_3\left(1+n\right)-\sqrt{p_1^2p_2^2p_3^2\left(1+n\right)^2-4p_1p_2p_3\beta\left(p_3-\beta\right)\left(p_2+p_1n^2\right)}}{2p_3\left(p_2+p_1n^2\right)},\quad \beta<\frac{p_3}{2}. \tag{23}$$

In this paper, two important cases will be investigated. First, we analsze the first case where the hopping rate of each chain is equal to p, (i.e., $p_1=p_2=p_3=p$). In this case, Eqs. (19)–(23) can be simplified as follows: for the (HD/HD/MC) phase,

$$\alpha_1>\frac{2-\sqrt{2}}{4n}p,\quad \beta>\frac{p}{2}; \tag{24}$$

for the (HD/HD/HD) phase,

$$\alpha_1>\frac{p-\sqrt{p^2-2\beta\left(p-\beta\right)}}{2n},\quad \beta<\frac{p}{2}; \tag{25}$$

for the (HD/LD/MC) phase,

$$\frac{n+1-\sqrt{2n}}{2\left(1+n^2\right)}p<\alpha_1<\frac{2-\sqrt{2}}{4n}p,\quad \beta>\frac{p}{2}; \tag{26}$$

for the (HD/LD/HD) phase,

$$\frac{p\left(n+1\right)-\sqrt{p^2\left(1+n\right)^2-4\beta\left(1+n^2\right)\left(p-\beta\right)}}{2\left(1+n^2\right)}<\alpha_1<\frac{p-\sqrt{p^2-2\beta\left(p-\beta\right)}}{2n},\quad \beta<\frac{p}{2}; \tag{27}$$

for the (LD/LD/LD) phase,

$$\alpha_1<\frac{n+1-\sqrt{2n}}{2\left(1+n^2\right)}p,\quad \beta>\frac{p}{2};$$
$$\alpha_1<\frac{p\left(n+1\right)-\sqrt{p^2\left(1+n\right)^2-4\beta\left(1+n^2\right)\left(p-\beta\right)}}{2\left(1+n^2\right)},\quad \beta<\frac{p}{2}. \tag{28}$$

When parameter n is fixed (i.e., $n=0.5$), with the decrease of hopping rate p (i.e., $p=1.0$, 0.8, 0.4 and 0.2, respectively), the HD/HD/MC phase is the only stationary state of which the region increases (see Figs. 2 and 5). It implies that when hopping rate reduces the system easily enters into the HD phase. When n is small enough, the HD/HD/MC phase also vanishes (see Fig. 3), and with the decrease of hopping rate p (i.e., $p=1.0$, 0.8, 0.4 and 0.2, sequentially), the HD/LD/MC phase is the only one of which the region increases (see Figs. 3 and 6). The relationship between parameters p and n are shown in Fig. 4. We can find that in the region which is under the tilt line the HD/HD/MC phase vanishes. The resulting density profiles are shown in Figs. 5 and 6. It is found that when the system is in the (HD/HD/MC) phase, the density profiles are independent of hopping rate (see Fig. 5(e)).

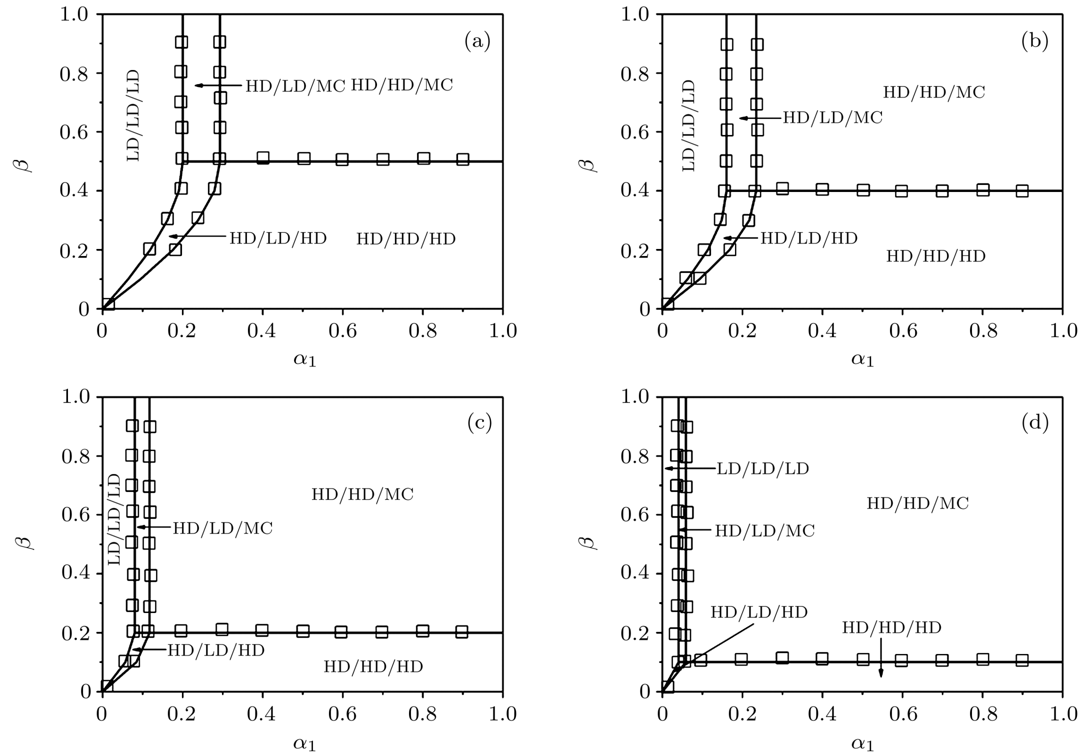


Fig. 2. Phase diagrams of the system with fixed parameter n (i.e., $n = 0.5$) and $p = 1.0$ (a), 0.8 (b), 0.4 (c) and 0.2 (d). Symbols denote the Monte Carlo simulations, while lines represent meal-field calculations.

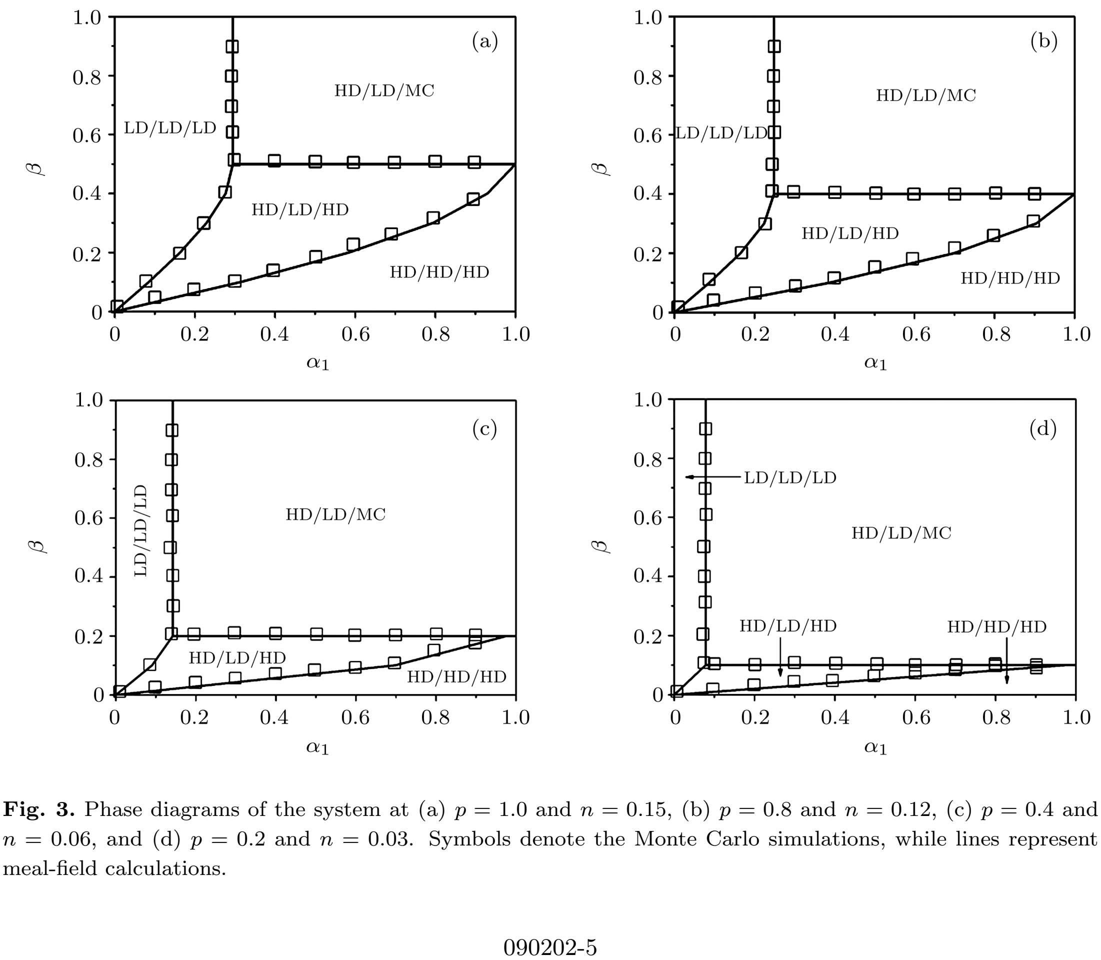


Fig. 3. Phase diagrams of the system at (a) $p = 1.0$ and $n = 0.15$, (b) $p = 0.8$ and $n = 0.12$, (c) $p = 0.4$ and $n = 0.06$, and (d) $p = 0.2$ and $n = 0.03$. Symbols denote the Monte Carlo simulations, while lines represent meal-field calculations.

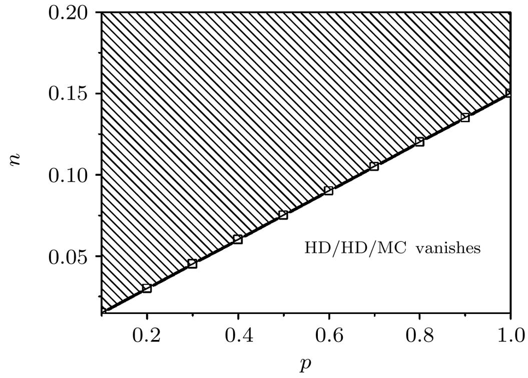


Fig. 4. Relationship between parameters p and n.

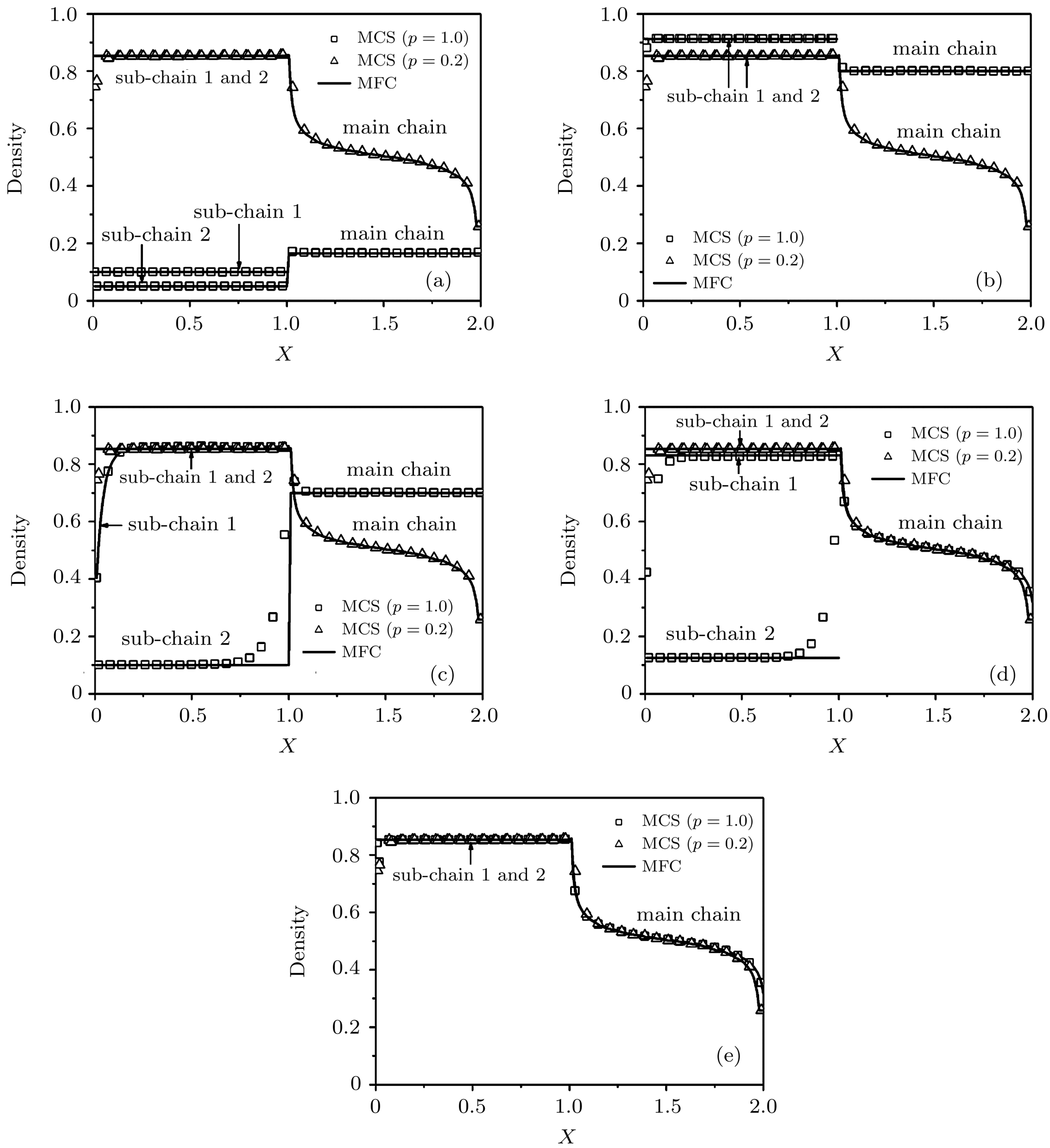


Fig. 5. Density profiles with parameter $n = 0.5$ for (a) $\alpha_1 = 0.1$, $\alpha_2 = 0.05$, $\beta = 0.8$; (b) $\alpha_1 = 0.8$, $\alpha_2 = 0.4$, $\beta = 0.2$; (c) $\alpha_1 = 0.21$, $\alpha_2 = 0.1$, $\beta = 0.3$; (d) $\alpha_1 = 0.25$, $\alpha_2 = 0.125$, $\beta = 0.8$; (e) $\alpha_1 = 0.8$, $\alpha_2 = 0.4$, $\beta = 0.8$. Symbols denote the Monte Carlo simulations while lines represent our theoretical predictions.

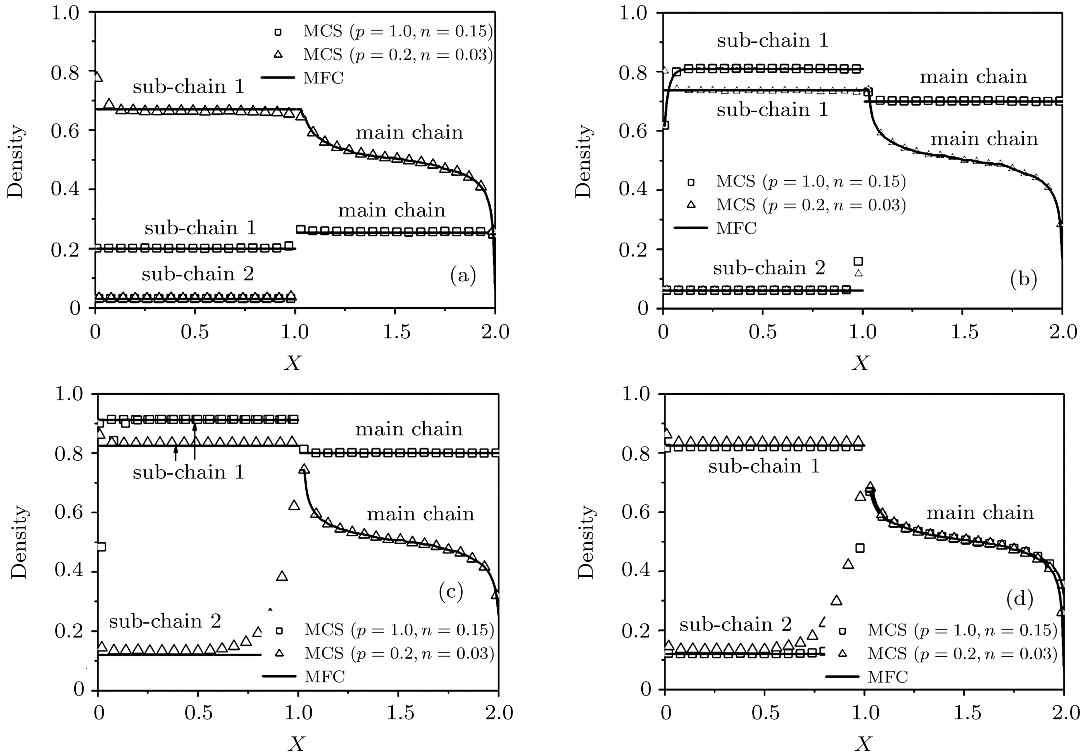


Fig. 6. Density profiles of the system for (a) $\alpha_1 = 0.2$, $\beta = 0.8$; (b) $\alpha_1 = 0.4$, $\beta = 0.3$; (c) $\alpha_1 = 0.8$, $\beta = 0.2$; (d) $\alpha_1 = 0.8$, $\beta = 0.8$. Symbols denote the Monte Carlo simulations while lines represent our theoretical predictions.

In the other case, the hopping rates of sub-chains are different from each other and the hopping rate of main chain is equal to one of sub-chains which may correspond to the traffic system with a passing lane. Assume that the hopping rates of sub-chain 1 and main chain are equal (i.e., $p_1 = p_3$); and the hopping rate sub-chain 2 is equal to p, we study the case where $p > p_1 = p_3$ (i.e., $p_1 = p_3 = mp$ and $m < 1$). Equations (19)–(23) can be simplified as follows: for the (HD/HD/MC) phase,

$$\alpha_1 > \frac{2-\sqrt{3-m}}{2n\,(m+1)}mp, \quad \beta > \frac{mp}{2}; \tag{29}$$

for the (HD/HD/HD) phase,

$$\frac{mp-\sqrt{m^2p^2-(m+1)\,\beta\,(mp-\beta)}}{n\,(m+1)} < \alpha_1, \quad \beta < \frac{mp}{2}; \tag{30}$$

for the (HD/LD/MC) phase,

$$\frac{(1+n)-\sqrt{(1-m)\,n^2+2n}}{2\,(1+mn^2)}mp < \alpha_1 < \frac{2-\sqrt{3-m}}{2n\,(m+1)}mp, \quad \beta > \frac{mp}{2}; \tag{31}$$

for the (HD/LD/HD) phase,

$$\begin{aligned} &\frac{mp\,(1+n)-\sqrt{m^2p^2\,(1+n)^2-4\beta\,(mp-\beta)\,(1+mn^2)}}{2\,(1+mn^2)} < \alpha_1, \\ &\alpha_1 < \frac{mp-\sqrt{m^2p^2-(m+1)\,\beta\,(mp-\beta)}}{n\,(m+1)}, \quad \beta < \frac{mp}{2}; \end{aligned} \tag{32}$$

for the (LD/LD/LD) phase,

$$\begin{aligned} &\alpha_1 < \frac{(1+n)-\sqrt{(1-m)\,n^2+2n}}{(1+mn^2)}mp, \quad \beta > \frac{mp}{2}; \\ &\alpha_1 < \frac{mp\,(1+n)-\sqrt{m^2p^2\,(1+n)^2-4\beta\,(mp-\beta)\,(1+mn^2)}}{2\,(1+mn^2)}, \quad \beta < \frac{mp}{2}. \end{aligned} \tag{33}$$

When parameter m is equal to 1 (i.e., $m = 1$), Eqs. (29)–(33) can be simplified into the first case where the hopping rates of chains are all equal to p.

When the parameter n is fixed (i.e., $n = 0.5$), with the synchronous decrease of hopping rate p and parameter m (i.e., $p(m) = 0.8$, 0.4 and 0.2, sequentially), the HD/HD/MC phase is the only stationary state of which the region increases sharply (see Figs. 7 and 10). It implies that when hopping rate and parameter m synchronously reduce the system easily enters into the HD phase. When n is extremely small, the HD/HD/MC phase also vanishes (see Fig. 8), and with the decrease of hopping rate p and parameter m (i.e., $p = 0.8$ and 0.4, $m = 0.8$ and 0.4 sequentially), the relationship between parameters p and n are shown in Fig. 9. We can find that in the region which is under the tilt line the HD/HD/MC phase vanishes. The HD/LD/MC phase is the only stationary state of which the region increases sharply (see Figs. 8 and 11). The resulting density profiles are shown in Figs. 10 and 11.

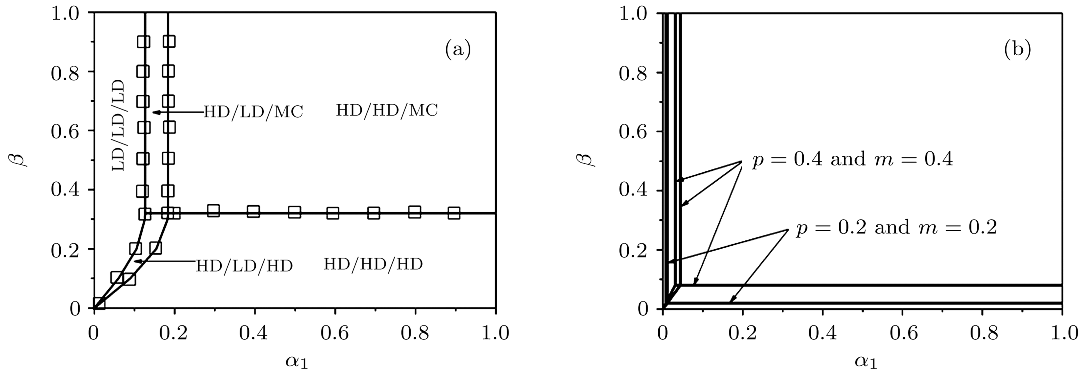


Fig. 7. Phase diagrams of the system with fixed parameter n (i.e., $n = 0.5$) and (a) $p = 0.8$, $m = 0.8$; (b) $p = 0.4$ and 0.2, $m = 0.4$ and 0.2. Symbols denote the Monte Carlo simulations while lines represent mean-field calculations.

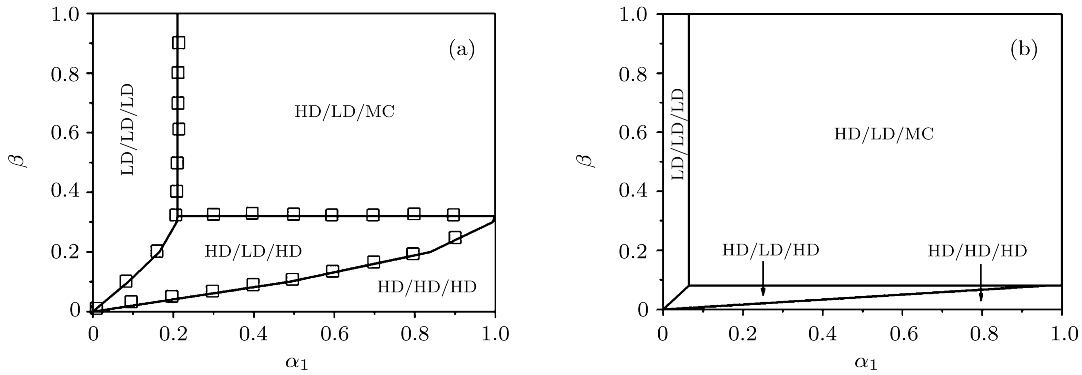


Fig. 8. Phase diagrams of the system with (a) $p = 0.8$, $m = 0.8$ and $n = 0.092$; (b) $p = 0.4$, $m = 0.4$ and $n = 0.023$. Symbols denote Monte Carlo simulations while lines represent meal-field calculations.

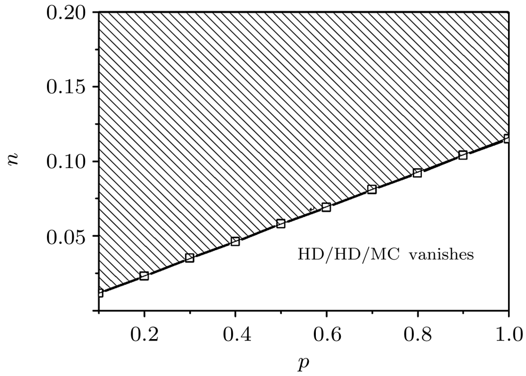


Fig. 9. Relationship between parameters p and n when m is fixed (i.e., $m = 0.8$).

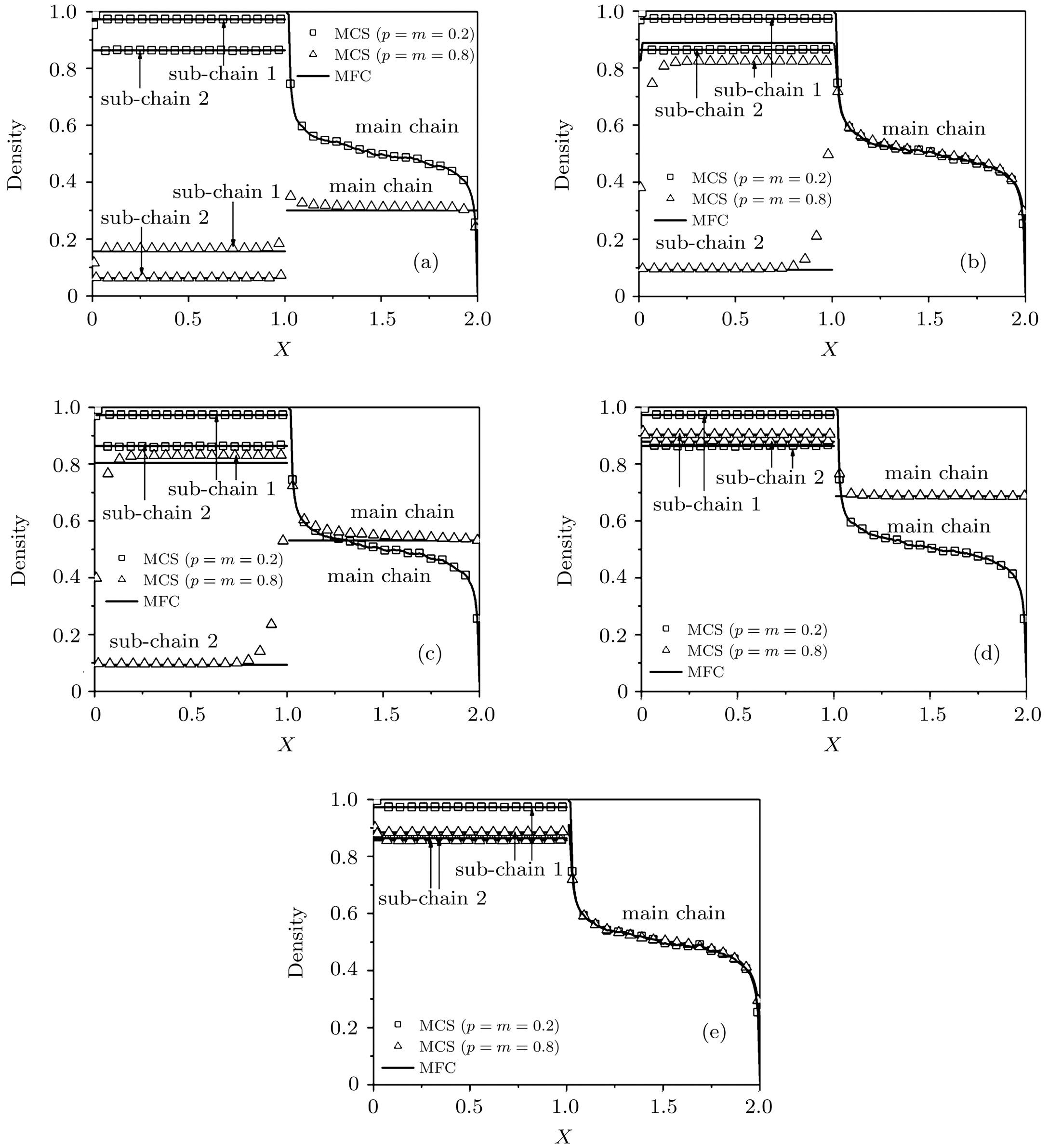
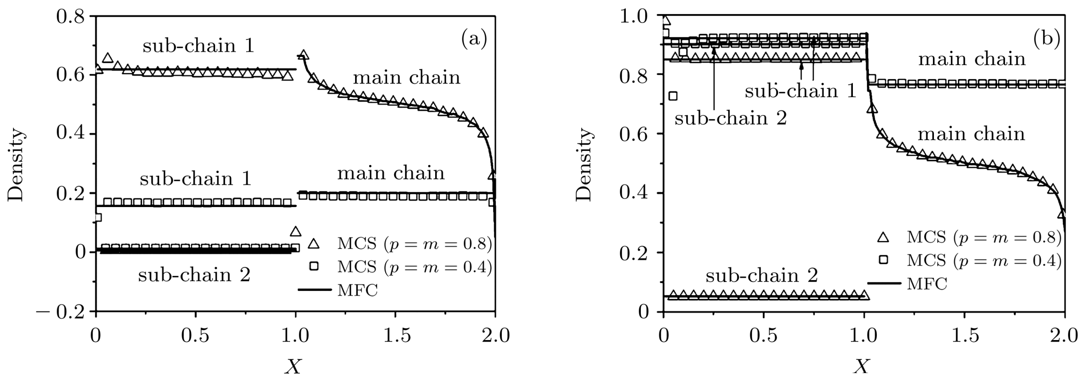


Fig. 10. Density profiles with parameter $n = 0.5$ and (a) $\alpha_1 = 0.1$, $\alpha_2 = 0.05$, $\beta = 0.8$; (b) $\alpha_1 = 0.15$, $\alpha_2 = 0.075$, $\beta = 0.8$; (c) $\alpha_1 = 0.15$, $\alpha_2 = 0.075$, $\beta = 0.3$; (d) $\alpha_1 = 0.8$, $\alpha_2 = 0.4$, $\beta = 0.2$; (e) $\alpha_1 = 0.8$, $\alpha_2 = 0.4$, $\beta = 0.8$. Symbols denote Monte Carlo simulations while lines represent our theoretical predictions.



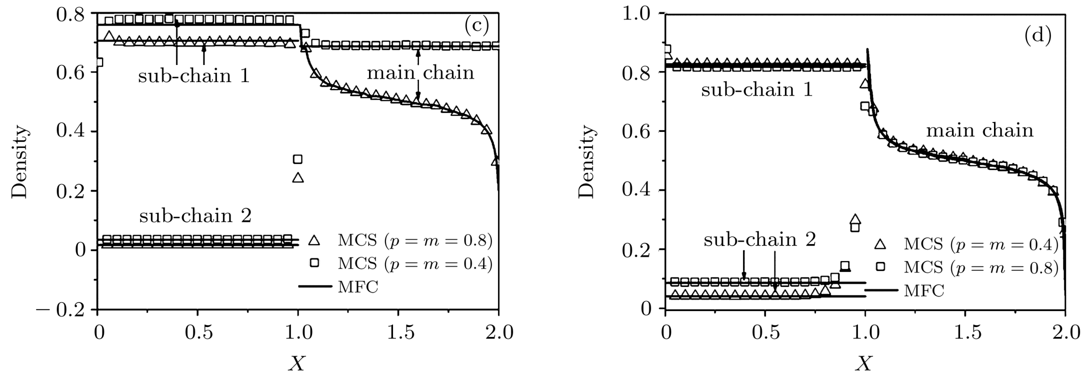


Fig. 11. Density profiles with parameter $n = 0.5$ and (a) $\alpha_1 = 0.1$, $\alpha_2 = 0.0023$, $\beta = 0.8$; (b) $\alpha_1 = 0.9$, $\alpha_2 = 0.0207$, $\beta = 0.15$; (c) $\alpha_1 = 0.3$, $\alpha_2 = 0.0069$, $\beta = 0.3$; (d) $\alpha_1 = 0.8$, $\alpha_2 = 0.0184$, $\beta = 0.8$. Symbols denote Monte Carlo simulations while lines represent our theoretical predictions.

3. Monte Carlo simulation

The theoretical results and the simulation results are discussed here. Theoretical results are verified by extensive Monte Carlo computer simulations. In simulations, we use $L = 1000$ for each of the chains and let the values of parameters α_1, α_2, p_1, p_2 and p_3 be varying. The stationary current profile and the stationary density profile can be obtained by sampling 1.1×10^9. The first 10^8 time steps are discarded as transients. Theoretical density profiles and the Monte Carlo computer simulations are shown in Figs. 5, 6, 10 and 11. We can find that they are consistent with each other.

From Fig. 2, we can find that when n is fixed (i.e., $n = 0.5$) and parameter p decreases from 1.0 to 0.2, the vertical boundary shifts toward the left and the horizontal boundary shifts toward the downside, namely, the region of the (HD/HD/MC) phase is the only region which increases. The (LD/LD/LD), the (HD/LD/MC), the (HD/LD/HD) and the (HD/HD/HD) phases change into the (HD/HD/MC) phase (see Figs. 5(a)–5(d)). Similarly, in Fig. 3, with parameters n and p decreasing, the vertical boundary shifts toward the left and the horizontal boundary shifts toward the downside, too. Namely, the region of the (HD/LD/MC) phase is the only region which increases. The (LD/LD/LD) phase, the (HD/LD/HD) phase and the (HD/HD/HD) phase change into the (HD/LD/MC) phase (see Figs. 6(a)–6(c)).

Similarly, in Fig. 7 the parameter n is fixed, the values of p and m decrease from 0.8 to 0.2. The vertical boundary shifts toward the left and the horizontal boundary shifts toward the downside, namely, the region of the (HD/HD/MC) phase is the only region which increases. The (LD/LD/LD) phase, the (HD/LD/MC) phase, the (HD/LD/HD) phase, and the (HD/HD/HD) phase change into the (HD/HD/MC) phase (see Figs. 10(a)–10(d)). In Fig. 8, with the values of p, m and n decreasing, the vertical boundary shifts toward the left and the horizontal boundary shifts toward the downside, too. The transition from the (LD/LD/LD) phase, the (HD/LD/HD) phase and the (HD/HD/HD) phase to the (HD/LD/MC) phase can be obtained (see Figs. 11(a)–11(c)).

4. Summary and conclusions

The effects of unequal injection rates and different hopping rates on asymmetric exclusion processes are investigated in this paper. The stationary-state behaviour of the system is studied by the mean-field approach. There are five stationary phases in the system, i.e., (LD/LD/LD), (HD/LD/MC), (HD/LD/HD), (HD/HD/HD) and (HD/HD/MC) phases. It is interesting that when parameter n is small enough, the HD/HD/MC phase vanishes.

In the first case (i.e., $p_1 = p_2 = p_3 = p$), when n is fixed (i.e., $n = 0.5$) and with the decrease of parameter p or with the decrease of parameters n and p, the vertical boundary shifts toward the left and the horizontal boundary shifts toward the downside, namely, the region of the (HD/HD/MC) or (HD/LD/MC) phase is the only region which increases.

Similarly, in another case ((i.e., $p_1 = p_3$, $p_2 = p$), when n is fixed (i.e., $n = 0.5$) and with the decrease of parameters p and n or with the decrease of parameters n, m and p, the vertical boundary shifts toward the left and the horizontal boundary shifts toward the downside, too. Namely, the region of the (HD/HD/MC) or (HD/LD/MC) phase is the only region which increases.

The density profiles are simulated and show that they are consistent well with the theoretical analyses. The approach used in this paper can be used directly to investigate the same model with parallel update.

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